

2.3 Machine Learning vs. Deep Learning

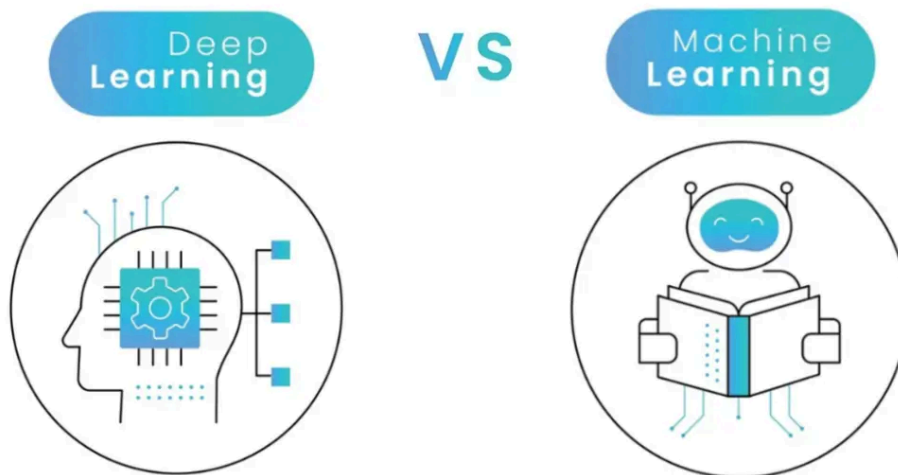
Kinjalben Desai

Department of AI and Data, Indiana Wesleyan University

(3SU2026) AIML-500: Machine Learning Fundamentals

Professor Antonio Singleton

July 19th, 2026



Machine Learning vs. Deep Learning: Real-World Applications

Introduction

AI is arguably the most impactful technology of the contemporary world, influencing industries including healthcare, finance, education, transportation, and retail. AI can be used by companies to automate tasks, analyze big volumes of data, enhance user experience, spot fraudulent activities, suggest relevant products, and facilitate decision-making processes. There are two major types of AI:

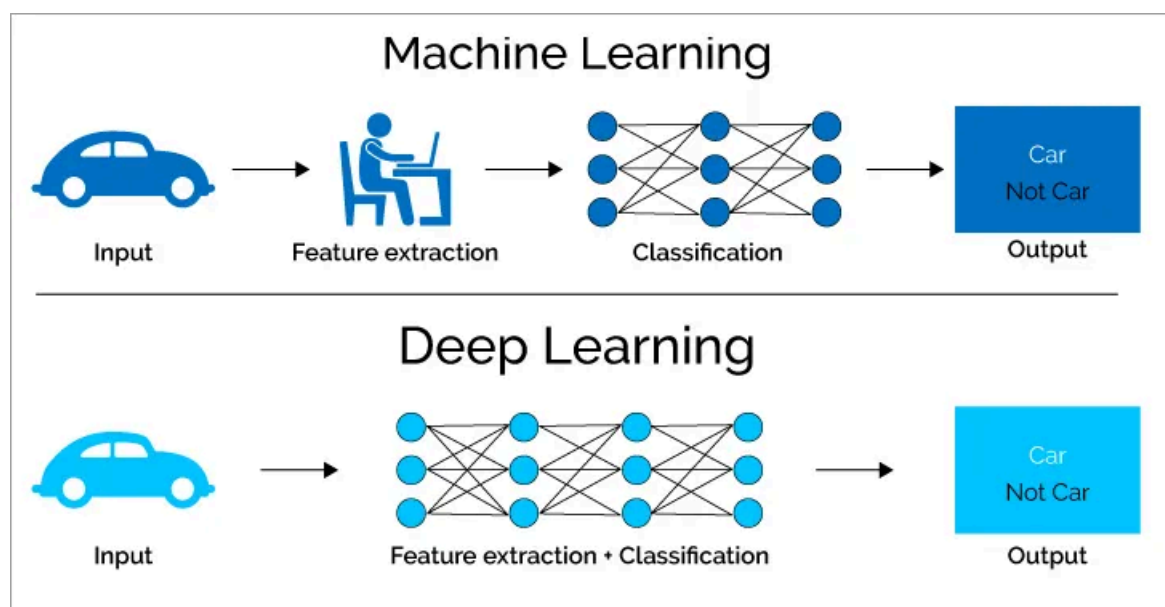
1. Machine Learning (ML)
2. Deep Learning (DL)

Even though both allow machines to learn from data and predict without direct programming, there are differences between ML and DL regarding the way data is processed, the volume of data that needs to be analyzed, and the level of difficulty of the problem.

"AI is probably the most important thing humanity has ever worked on. I think of it as something more profound than electricity or fire."

- Sundar Pichai, CEO of Google and Alphabet Inc.

The quote from Sundar Pichai indicates the immense role that is envisioned to be played by artificial intelligence in society. Just like the role played by electricity in transforming the manner in which life and work was conducted, AI is transforming business and other activities through increased efficiency and intelligence. Machine learning and deep learning form the basis of many of the AI algorithms. Both machine learning and deep learning algorithms allow computers to learn about the environment and make intelligent decisions. However, there are differences between the two approaches in terms of features, data, and computations.



Machine Learning Example: Email Spam Detection

An example of machine learning covered in the lesson is email spam classification. Services like Gmail and Microsoft Outlook utilize machine learning to classify whether an email should be added to the user's inbox or to the spam box. Logistic Regression and Decision Trees algorithms analyze predetermined features such as suspicious words used, sender credibility, the amount of hyperlinks, attachments and previous spam complaints. With the use of thousands of samples of spam and legitimate emails, the algorithm will be able to properly classify new emails.

Machine learning is the right approach to solve this problem since all the key features that define a spam email are already known and can be selected prior to modeling. Additionally, the data used is rather structured which makes the algorithm able to predict results using less resources compared to the deep learning technique. One more advantage is that the models of machine learning are easier to train and modify in case of new spam trends.

Deep learning is not ideal for simple spam detection since it generally needs bigger data sets and stronger computers with more time to train. Even though deep learning can recognize more advanced patterns in more elaborate phishing schemes, such complexity is unnecessary in simple spam detection since machine learning works perfectly well.

Deep Learning Example: Image Recognition

One deep learning example discussed in the class is the application of **Convolutional Neural Networks (CNNs)** in the field of image recognition. An example of an application of this concept is the use of facial recognition in smartphones like Face ID, airport security and in social media platforms which automatically recognize and label people in pictures.

Deep learning is the way to go due to the high complexity of images that have millions of pixels. Unlike in traditional machine learning where one needs to specify the features of interest (e.g., shape of eye, proportionality of face or skin textures), deep learning automatically identifies such features from raw data through the CNN. The system will be able to identify images of faces even when the lighting conditions vary or the angle of view changes since it will first identify basic patterns such as edges and then recognize more complex facial patterns.

Traditional machine learning does not work well since it involves the specification of the image features prior to training the model. Due to the complexity of images, feature identification becomes time consuming and also less accurate compared to deep learning.

Key Differences Between Machine Learning and Deep Learning

Machine learning and deep learning have some fundamental differences despite being similar to each other.

Feature Extraction: Human intervention is required to select the right features in case of machine learning. Deep learning extracts features automatically from raw data itself using many layers of neural networks.

Data Requirement: Machine learning works well on small amounts of data along with structured data. Deep learning, on the other hand, requires huge data in order to perform accurately as it does not require manual selection of features.

Complexity: The machine learning models are simple to understand and easy to implement while deep learning models consist of many layers of neurons and parameters.

An example that can help distinguish machine learning from deep learning using an analogy from the lesson is as follows: **Machine learning would be like having an email checklist to determine whether an email is spam while deep learning would be like teaching the student how to read and then finding out the features of spam emails.**

Conclusion

Machine Learning and Deep Learning are two highly advanced technologies that keep changing industries across the world. However, each technology is applicable in solving particular problems. In the case of machine learning, this technology should be preferred to solve structured data problems where features are well-defined, like detecting email spam. This technology is considered efficient and accurate and does not require many computing powers. As for deep learning, it is best suited for solving complicated problems like recognizing images because it allows automatically learning patterns within data. The comprehension of the strengths and weaknesses of both technologies helps organizations to choose the most appropriate AI technology.

References

- AIML. (2024). *Machine Learning Resources*. <https://aiml.com/>
- AIMultiple. (2024). *Top Deep Learning Use Cases & Examples in 2024*. <https://aimultiple.com/deep-learning-applications>
- Google Cloud. (n.d.). *What is machine learning?* <https://cloud.google.com/learn/what-is-machine-learning>
- IBM. (n.d.). *What is deep learning?* <https://www.ibm.com/topics/deep-learning>
- Pichai, S. (2018). *AI is probably the most important thing humanity has ever worked on*. World Economic Forum. <https://www.weforum.org/stories/2018/11/artificial-intelligence-ai-sundar-pichai-google/>